Chaiyong Mun

Email: chaiyong98@snu.ac.kr
GitHub: github.com/chaiyong0401

Education

Seoul National University Seoul, South Korea
Integrated M.S./Ph.D. program in the Department of Intelligence and Information. 2023 Fall–Current
- GPA: 4.03/4.3

Seoul National University Seoul, South Korea
B.S. in Electrical and Computer Engineering 2017–2023 Fall
- GPA: 3.5/4.3

Experience

Seoul National University/DYROS Lab. Seoul, South Korea
Research Assistant/Dynamic Robotic Systems Laboratory 2024–2025
- Development of humanoid robots that feel like humans, communicate, and grow through learning with MOTIE.
- Imitation learning that plans the target position and force of the end-effector using visual and tactile information.

Publications

[1] **C. Mun**, E. Sung, H. Lee, and J. Park, "Hierarchical reactive grasping for contact-rich manipulation via topological visuo-tactile learning", *Accepted for Publication at IEEE International Conference on Intelligent Robots and Systems (IROS) 2026*, 2026.

[2] M. Suh, **C. Mun**, H. Lee, and J. Park, "Grasp state estimation via 3d mean resultant length", *Proceedings of the Korea Robotics Society Annual Conference (KRoC)*, 2026.

[3] E. Sung, S. You, J. Shin, S. Park, S. Hur, J. Kim, S. Byun, **C. Mun**, H. Lee, D. Kim, and J. Park, "Snu-avatar haptic arm: A hybrid qdd/dd operator interface for teleoperation", *IEEE Access*, vol. 14, pp. 57 409–57 426, 2026.

Research Outputs

Journal Manuscript in Preparation
One-Class Tactile Failure Detection and Recovery for Autonomous Laundry Handling.

Patent Application in Preparation
Method and Apparatus for Robot Grasp State Estimation Using Dual-Channel Tactile Anomaly Detection with Mean Resultant Length-Based Features.

Teaching

- **Teaching Assistant** at Seoul National University 2025
  DYROS Robotics Boot Camp II (Robot Design, SI, Simulation and Control)
- **Teaching Assistant** at Seoul National University 2025
  DYROS Robotics Boot Camp I (ROS: Robot Operating System)

Skills

- **Software:** C++, Python, MATLAB, PyTorch, Anaconda, Docker, C
- **System Integration:** RTLinux, ROS1, ROS2, EtherCAT
- **Applications:** Isaac Gym/Lab, MuJoCo, CoppeliaSim, MoveIt, Git, VS code, Notion.

Languages

- **Language:** Korean (mother tongue)
- **Language:** English (intermediate)

Projects

AI-based Robot Control Algorithm Development (Samsung DA, Project Lead, 2026)
Developed an autonomous laundry handling system for domestic service robots by integrating visual perception, tactile sensing, and closed-loop failure recovery. The system employs domain-conditioned segmentation and one-class tactile anomaly detection to enable reliable manipulation of highly deformable garments.

Learning-based Humanoid Robot Manipulation (MOTIE, Research Assistant, 2024–2025)
Developed a learning-based robot manipulation framework that integrates visual and tactile imitation learning to generate end-effector pose and force commands for robust object manipulation.

Haptic Teleoperation for Humanoid Robot Manipulation (DYROS, Project Lead, 2024)
Developed a MuJoCo simulation environment for a laboratory humanoid robot and implemented haptic teleoperation for object grasping and manipulation.

Transparent Object Grasp Pose Estimation (Samsung SAIT, Research Assistant, 2023)
Developed a vision-based transparent object grasp pose estimation framework by integrating RGB-D depth restoration and NeRF-based perception for reliable robotic manipulation.

Research Interest

My research focuses on multimodal robot manipulation through visual–tactile perception and robot learning. Building on my experience in perception, tactile sensing, imitation learning, and autonomous manipulation, I aim to develop Vision-Language-Action (VLA) models that enable humanoid robots to understand, reason about, and interact with complex real-world environments.